

ACTIVE INFERENCE & HCI

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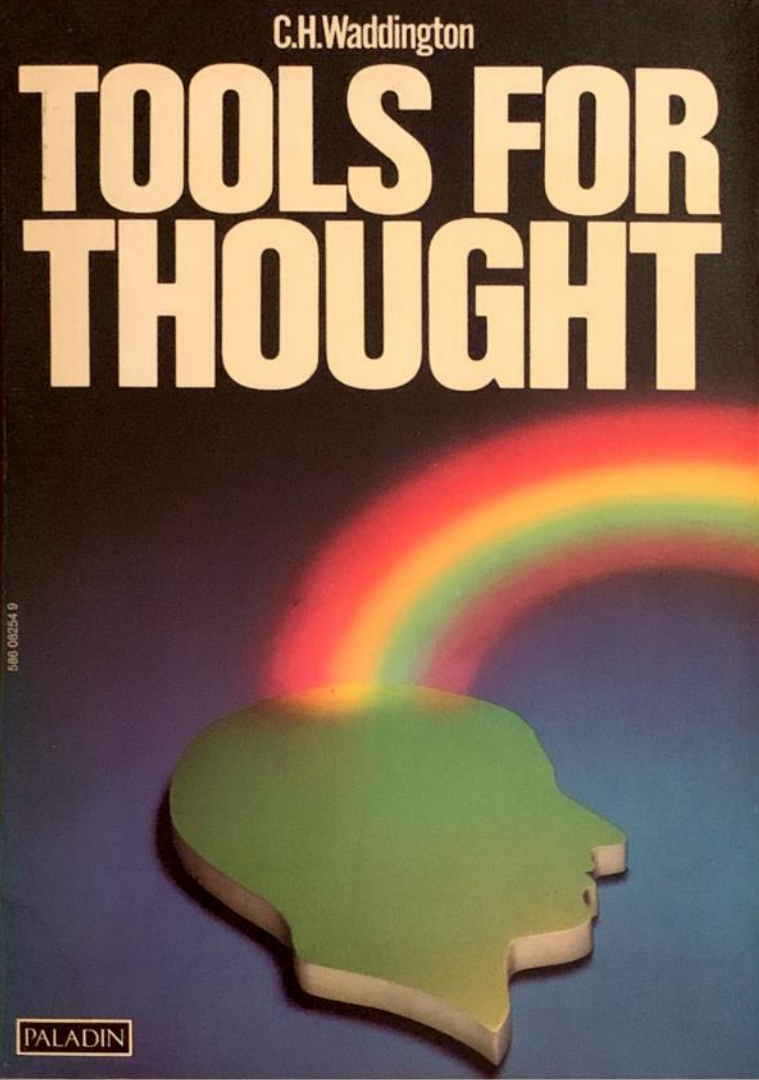
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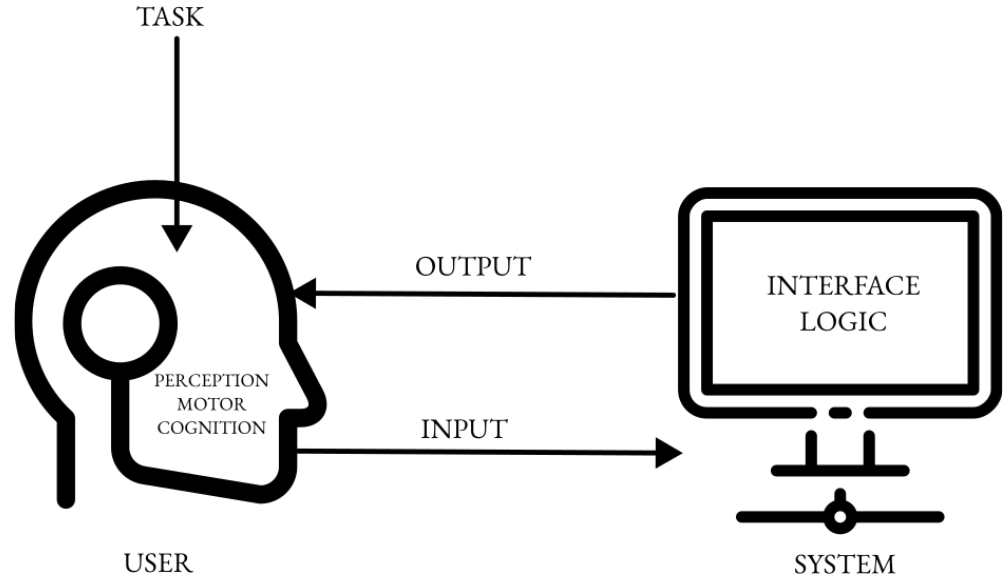


Wrong way down a one-way street

A user, viewed as an information processor, engages in a task, providing input to a computer and receiving output.

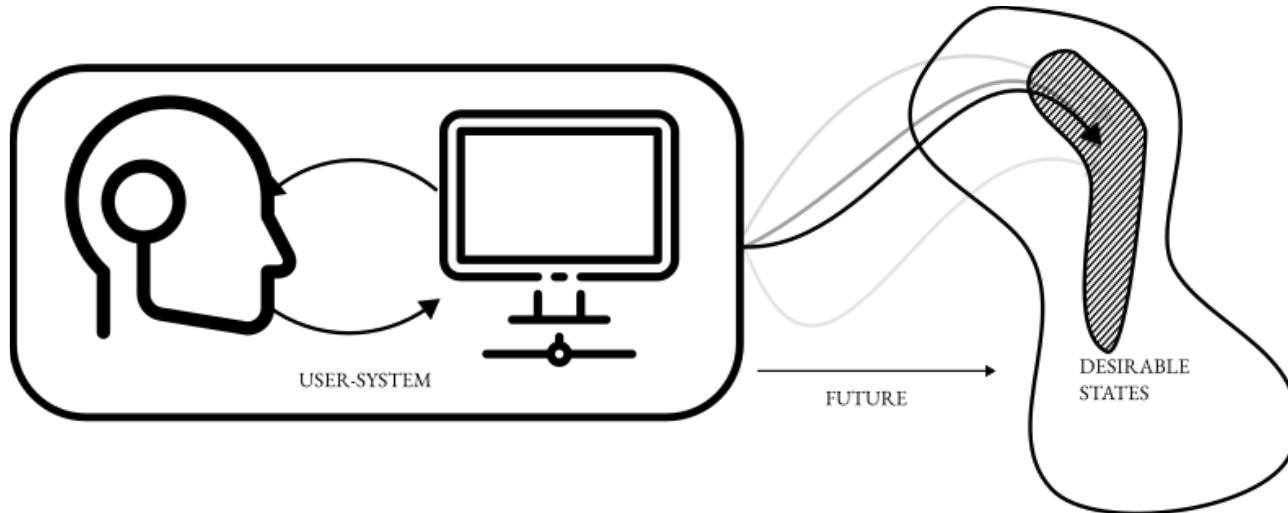
The interface, taken to be the input/output boundary of a system is designed and then evaluated and refined against metrics like time to task completion, satisfaction, learning time, etc.

Once refined, this interface is frozen and deployed to the world.

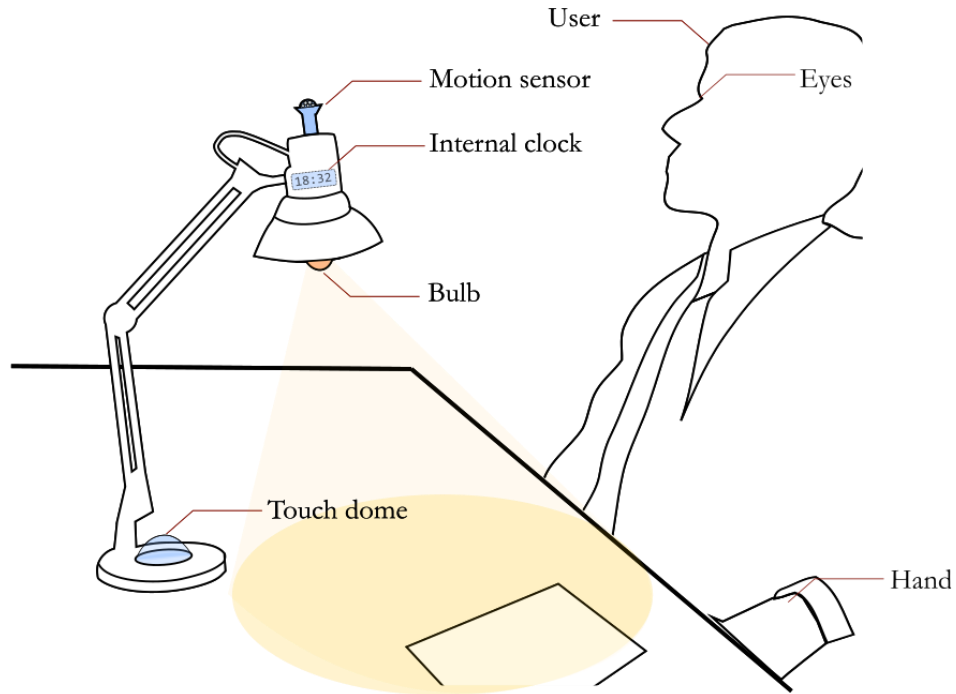


Active inference

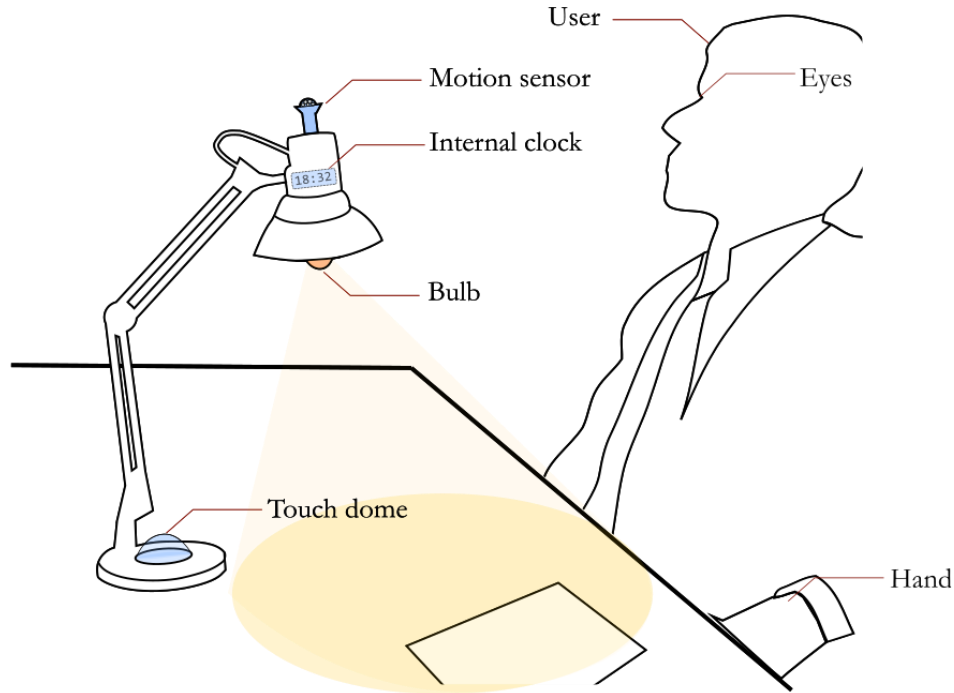
Active inference is a computational theory of interaction in which users are not exchanging inputs and outputs with a passive system; user and systems rather continually act as a joint system to make their future observations predictable and preferred: **co-regulation of future states of a combined sensorium.**



A lamp



A lamp

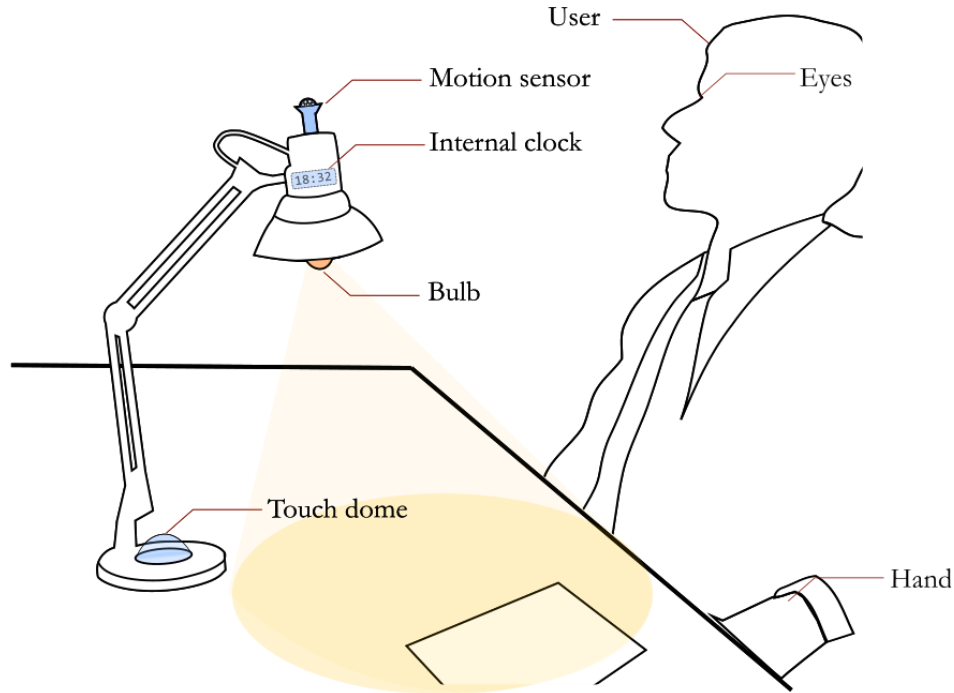


Classical HCI

The lamp is designed to support the user in the task of attaining suitable illumination via controls and feedback and refined via iterative evaluation. Design process considers time to learn, error rate, etc.

“is” but not “do”

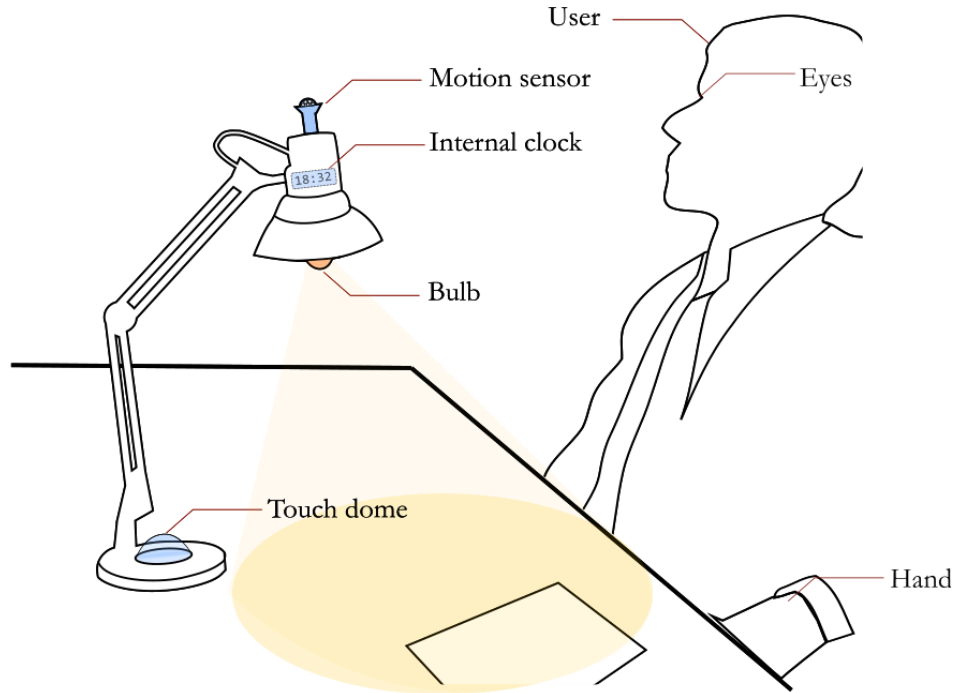
A lamp



Perceptual control

The user acts to hold the perceived illumination at a desired reference and the lamp affords control surfaces to support that. The feedback loop regulates illumination.

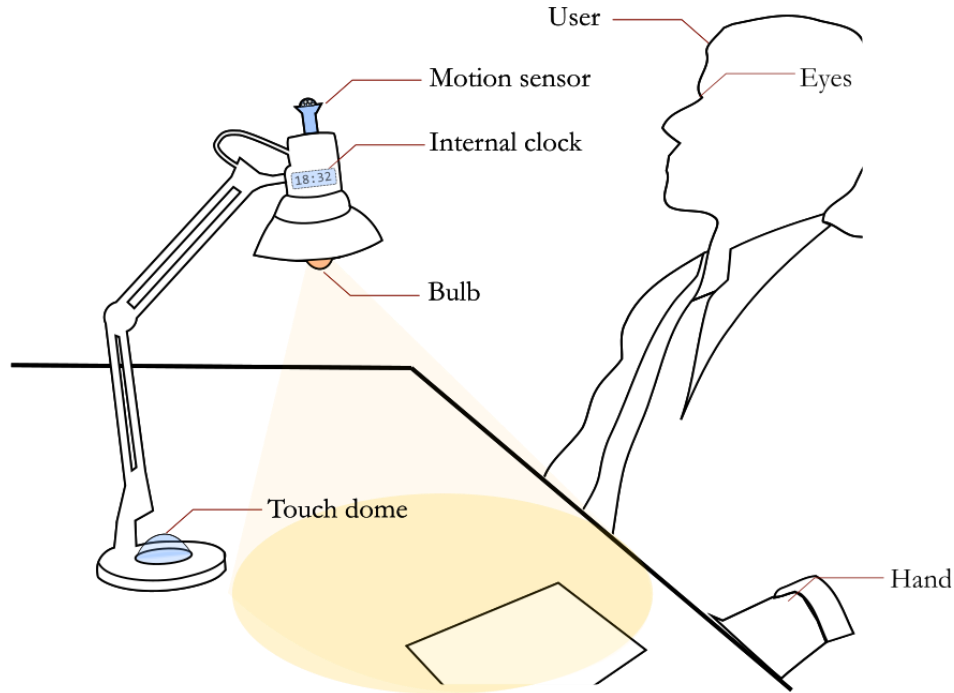
A lamp



Computational Rationality

The user is a bounded-optimal agent who, given noisy perception and imprecise control, adjusts the lamp to maximise the utility of being well-lit. The designer models that bounded-optimal behaviour and solves for the lamp's controls.

A lamp

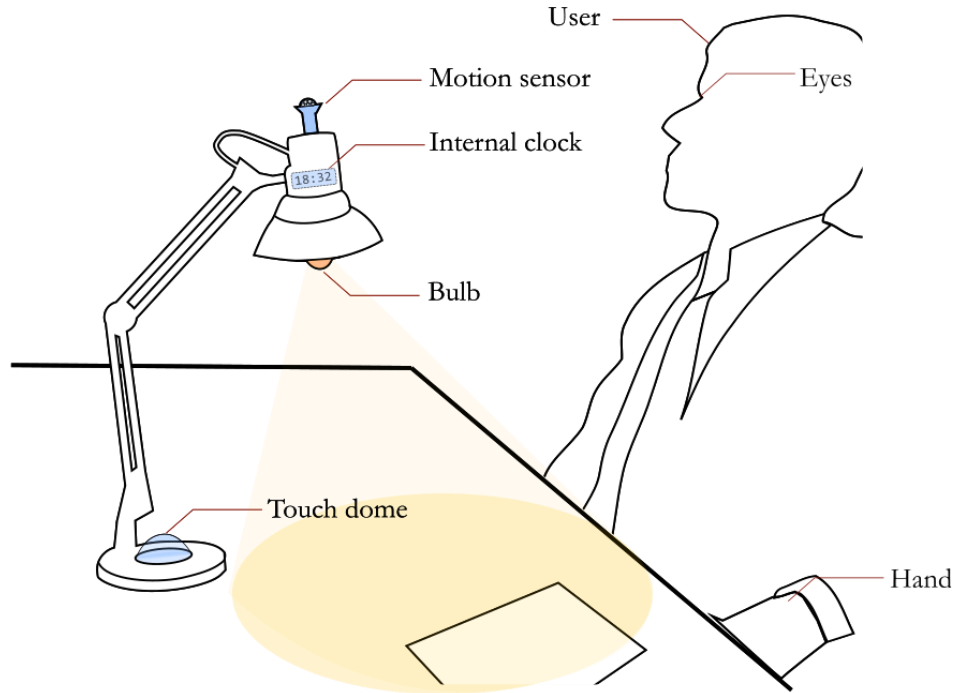


Reinforcement learning

The lamp learns responses to input that maximise a specified reward (“user likes this”), normally offline.

Learning creates an efficacious and potentially complex mapping between input and illumination.

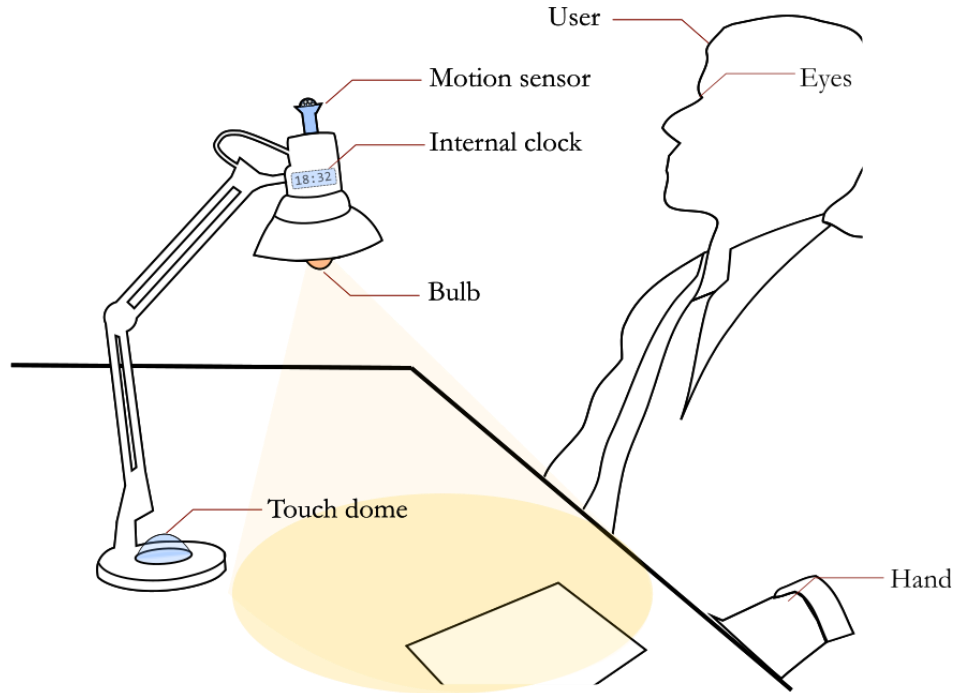
A lamp



Active inference

The lamp acts to minimise the long-term surprise of the joint human-lamp system; that is to maximise the probability of being in future preferable states.

A lamp



Active inference manifest

Joint, not servant

Anticipatory

Attentive (probing, tracking)

Graceful

Satiable

Contrasts

Classical HCI

Emphasises tasks, input/output and metrics. Design is foregrounded as the way to imbue a system with desirable properties.

Maximises specific metrics.

Perceptual Control

Emphasises **homeostasis** of percepts in the now. Hold an observation at a set-point by rejecting disturbances.

Minimises error.

Reinforcement learning

Emphasises maximising reward as a strategy to plan. Rewards are extrinsic and are associated to the environment.

Maximises reward.

Computational rationality

Emphasises bounded-optimal behaviour: the best a resource-limited agent can do at a task given cognitive/motor bounds. Scalar, extrinsic utility; user is the agent.

Maximises bounded-optimal utility.

Active Inference

Emphasises agents, not necessarily the user, minimising expected free energy of uncertain future states. EFE combines exploitation and exploration.

Minimises surprise*.

* “Surprise” idiosyncratically defined to mean unexpected *and* undesirable in the long-term.

An agent acts to observe the futures it would prefer to be in.

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An agent seeks futures that are evidence of its continued existence.

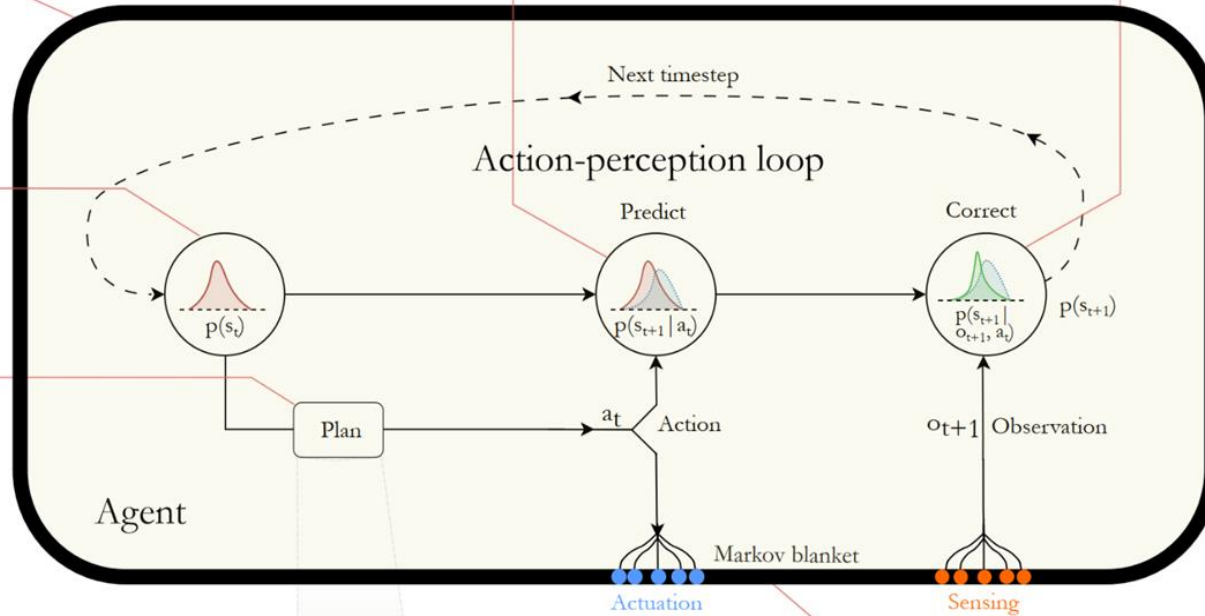
The environment is inaccessible to the agent, except via the **Markov blanket**.

Beliefs are always represented as distributions over possible configurations.

Planning involves searching policies -- action sequences -- scoring them and selecting a next action.

The agent uses a **forward model** to predict future states of the world compatible with its previous belief (and the selected action)

Observations from the environment are used to correct the belief to be compatible with what is sensed. This is a **Bayesian update**, posterior $\propto$ prior $\times$ evidence.



The **Markov blanket** captures all of the variables on the boundary between the agent and the environment.

Environment

Reasoning in the future perfect

“[Active Inference] turns the action selection problem on its head. Instead of saying: ‘*I have some goal, what do I have to do to achieve it?*’, the active inference agent asks: ‘*Given that my goals were achieved, what would have been the most probable actions that I took?*’”

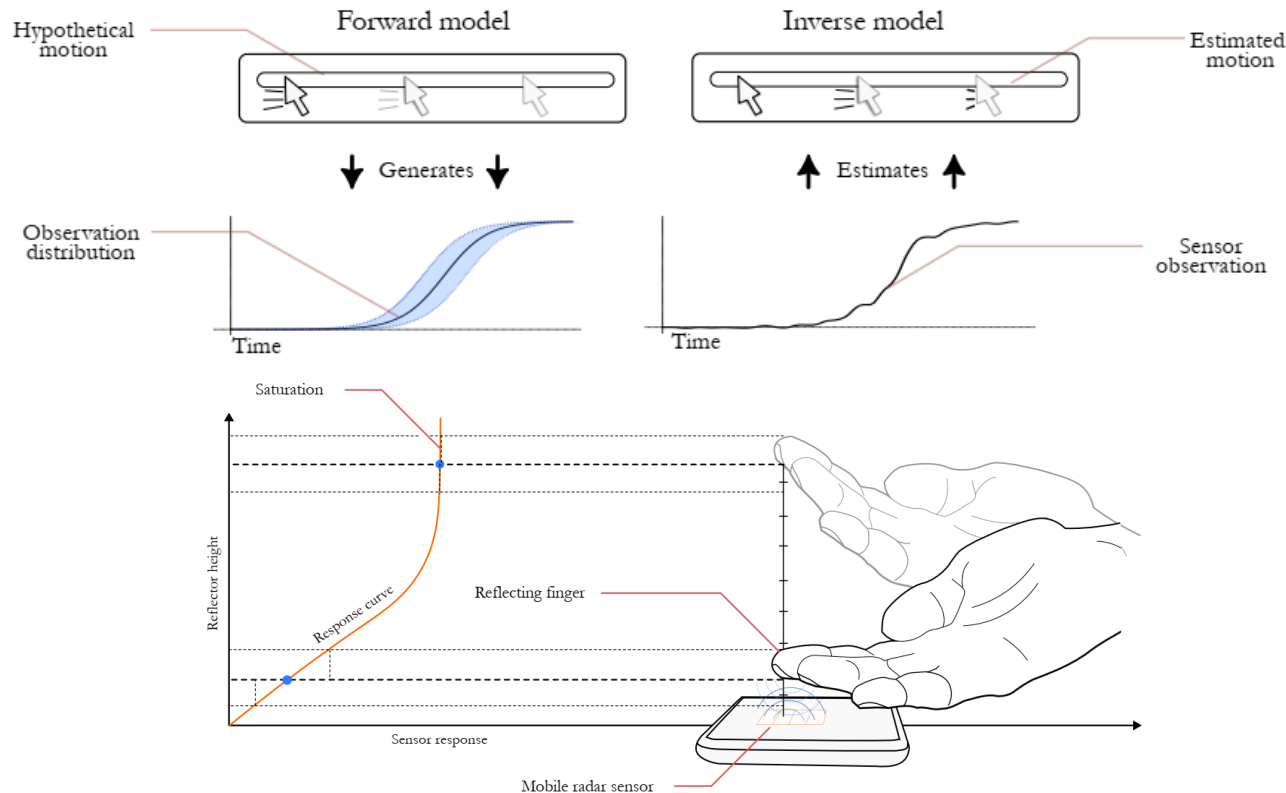
Millidge et al., “Whence the expected free energy”, Neural Computation, 2021



Forward, not inverse

Active inference treats all observations as the consequence of hidden states.

This emphasises modelling the dynamics of causes rather than the black-box inference of the implications of observations.



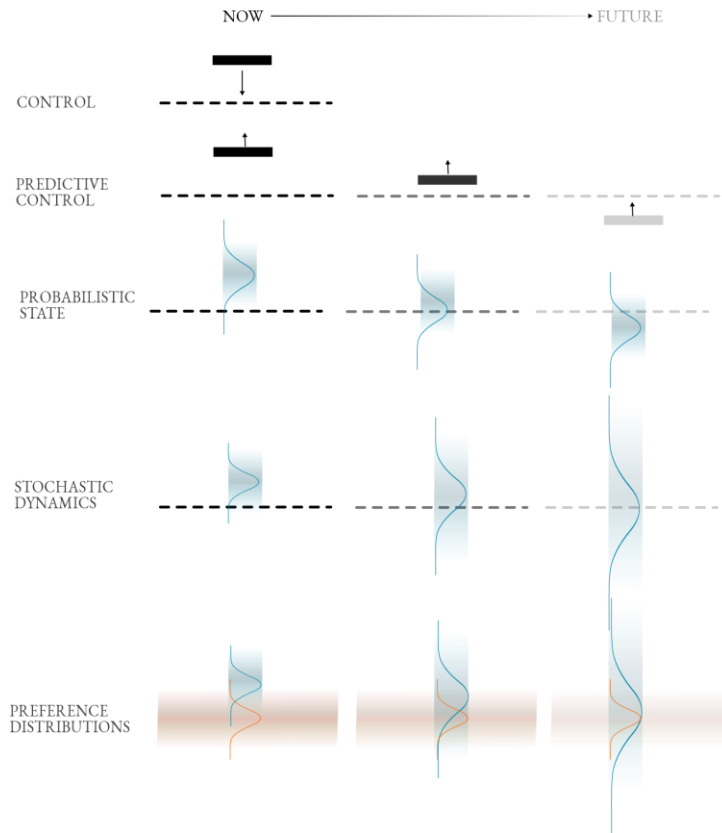
Prefer, don't aim

Active inference is exclusively probabilistic.

Everything is a distribution: beliefs, dynamics, observations, **and** preferences.

Updates are Bayesian in principle, approximate in practice.

We reason in **preference distributions**, not goals.



What futures are preferred now?

$$\text{EFE} = \underbrace{\text{expected cost} - \text{expected information gain}}_{\text{Expected free energy}}$$

What futures are preferred now?

$$\begin{aligned}
 \text{Expected Free Energy} \\
 G(\pi) = \sum_{\tau=t+1}^T & \left[\underbrace{\mathbb{E}_{q(o_\tau|\pi)} \left[\underbrace{-\ln p^*(o_\tau)}_{\substack{\text{cost assigned by} \\ \text{preference prior}}} \right]}_{\substack{\text{pragmatic} \\ \text{expected preference cost at } \tau}} - \underbrace{\mathbb{E}_{q(o_\tau|\pi)} D_{\text{KL}} \left(\underbrace{q(s_\tau | o_\tau, \pi)}_{\substack{\text{posterior belief} \\ \text{after possible observation}}} \parallel \underbrace{q(s_\tau | \pi)}_{\substack{\text{predicted belief} \\ \text{before observation}}} \right)}_{\substack{\text{information gained} \\ \text{about hidden state}}} \right] \\
 & \underbrace{\hspace{10em}}_{\substack{\text{epistemic} \\ \text{expected belief change at } \tau}} \\
 & \underbrace{\hspace{15em}}_{\substack{\text{sum over planning horizon} \\ \text{evaluate each future time } \tau=t+1, \dots, T}}
 \end{aligned}$$

$$\underbrace{q(o_\tau | \pi)}_{\substack{\text{distribution of observations} \\ \text{given policy } \pi \text{ at time } \tau}} = \int \underbrace{p(o_\tau | s_\tau)}_{\substack{\text{likelihood} \\ \text{how hidden states generate observations}}} \underbrace{q(s_\tau | \pi)}_{\substack{\text{predicted belief over hidden states} \\ \text{if policy } \pi \text{ is followed}}} ds_\tau$$

$$q(\pi) = \sigma \left(\underbrace{\begin{pmatrix} \text{negative expected free energy} \\ \text{scaled by policy precision } \gamma \\ -\gamma G(\pi) \end{pmatrix}}_{\substack{\text{posterior over policies} \\ \text{lower EFE policies become more probable}}} \right)$$

$$\underbrace{\pi \sim q(\pi)}_{\substack{\text{sample a policy} \\ \text{from the policy posterior}}} \Rightarrow$$

$$\underbrace{a_t = \pi(t)}_{\substack{\text{execute the first action} \\ \text{of the selected policy}}}$$

Who is U?

We often optimise interfaces as if there are users, tasks, cost functions, and design variables. However, modern interaction involves adaptive systems, uncertain sensing, partial autonomy, unsurfaced latent user goals, and co-adaptation.

This pivots the question from “**what is the optimal interface?**” to “**what kind of closed-loop being will we engineer?**”

But we can only engineer one half of the being directly!

What is E?

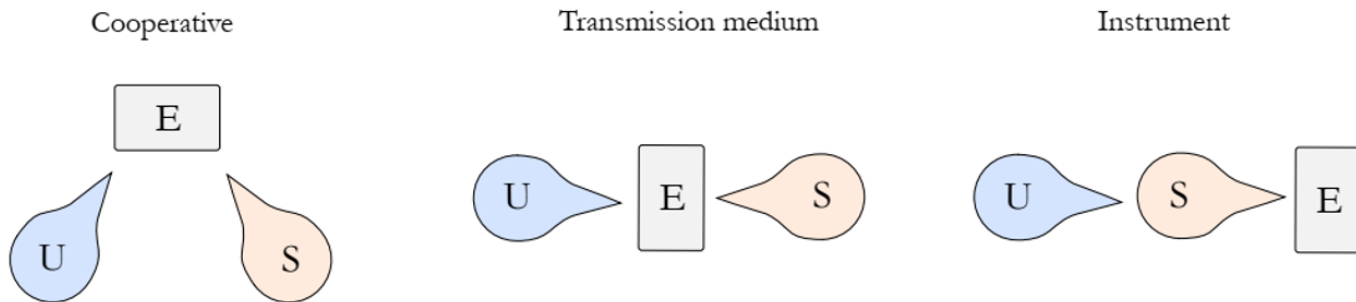
What is an agent and how is it distinguished from its environment (E)?

This is not as simple as identifying an agent with a user, or with an interface or system.

An agent's boundary is not a "skin bag".

Careful reasoning about the agent, its boundaries, and the environment is one of the critical perspective shifts in active inference modelling.

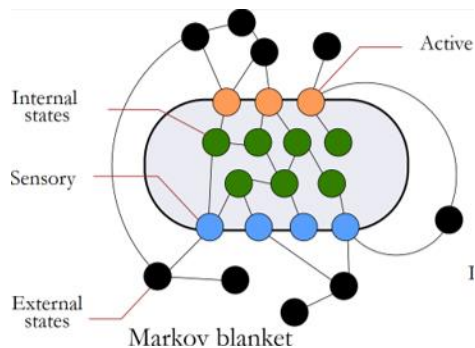
In HCI, there is a **dyad** of agents bound to a common environment.



Where the (inter)action is

HCI is perhaps the only field dedicated to a boundary (the interface) that it rarely formalises.

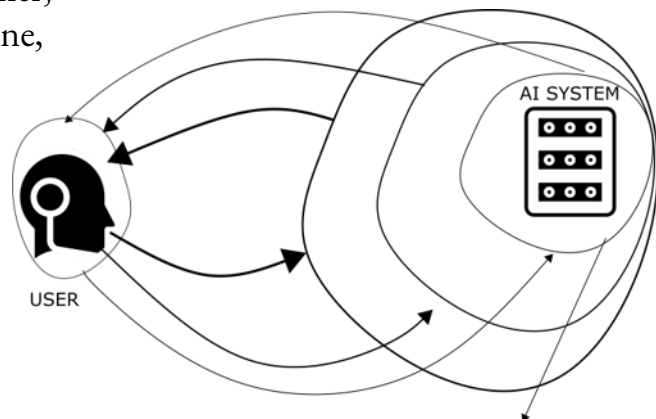
Active inference proposes a precise, statistical concept.



A Markov blanket is a set of boundary states of an agent and its environment, comprising actuation and observation.

Blankets are ephemeral and ebb and flow, just as one picks up a hammer, sits in a cockpit or touches a phone, thus becoming a different composite being.

This is a perspective that rationalises and unifies many key concepts in HCI: embodiment, agency, H-metaphor, shared-autonomy in a single computational concept.



A photograph of a sailboat on a blue sea with mountains in the background. The sailboat is on the right side of the frame, and its white sail is visible. The sea is a deep blue, and the mountains are in the distance under a clear sky.

When you are in the groove, the boat is you.

But the boat is you only while you keep
evidencing states that keep yourselves as one.

*“creatures like us... are nature's experts at knitting
their own Markov blankets... self-organizing
processes that constantly re-invent themselves,
repeatedly re-defining their own cognitive, bodily,
and sensory forms”*

**Andy Clark, "How to Knit Your Own Markov Blanket,"
Philosophy and Predictive Processing (2017)**

A useful notation

Brackets say who is doing the inferring,
 primes say who is being modelled,
 and nesting specifies theories of mind.

$U(S)$ – A user operating an active inference system

$(U')(S)$ – An AIF model of a user interacting with an AIF system

$U(S U')$ – An AIF system with a fixed model of a user

$U(S(U'))$ – A user interacting with an AIF system that has an AIF model of the user

$((U(S'))(S(U')))$ – A joint system, each with internal agents that model the other

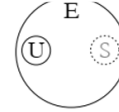
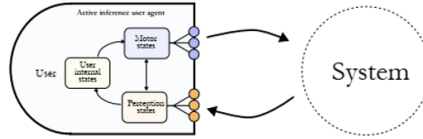
Offline

Active inference modeling used in the design and evaluation of systems via simulation of user behaviour.

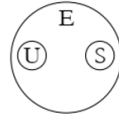
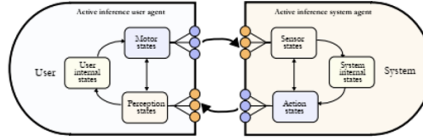
Level 1

1.

Description (U) S



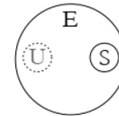
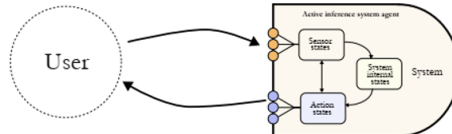
Mutual Interaction (U) (S)



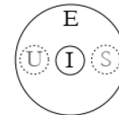
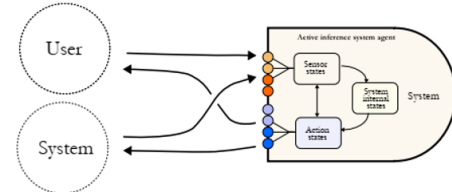
Online

Active inference "in the loop", applied at run-time to improve interaction.

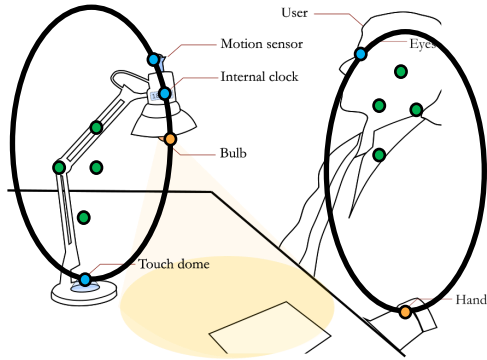
Construction U (S)



Transduction U (I) S

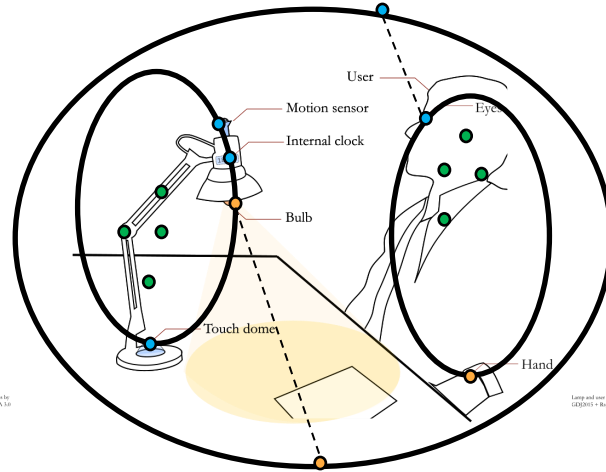


A lamp



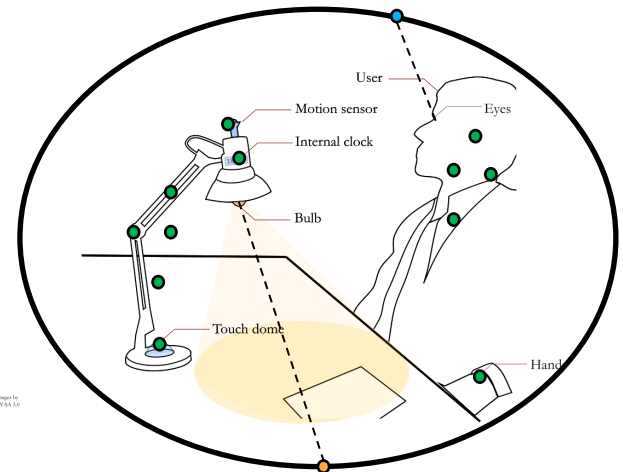
(U) (S)

I have a smart lamp, and it anticipates and elicits.



((U) (S))

I have a smart lamp,
and we regulate our environment.



(U S)

The lamp is part of me,
just as my iris is.

What does this buy us?

Model-based **operationalization** of questions around autonomy and agency.

Specification by defining preference, not optimizing goals

Unification: one currency -- expected free energy -- to recast perception, planning, and action into a single problem.

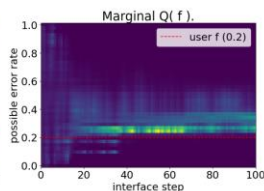
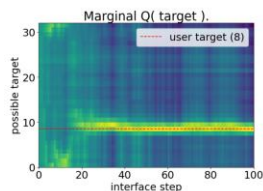
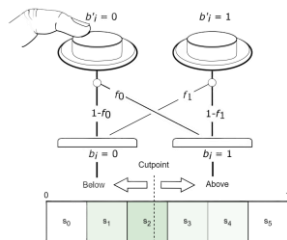
Computational procedures to realise a precise kind of intelligent interfaces: probabilistic (graceful, robust) and proactively anticipatory.

From button to limb

EFE w/o prediction

Stein et al. 2024

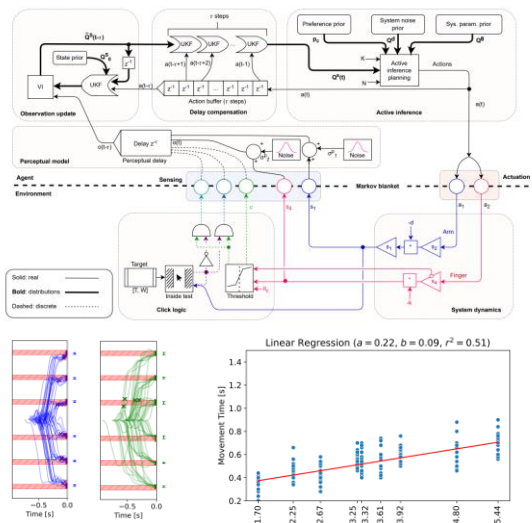
Showed an active inference agent can communicate reliably over a noisy channel, even when the noise is unknown.



Predictive models

Klar et al. 2025

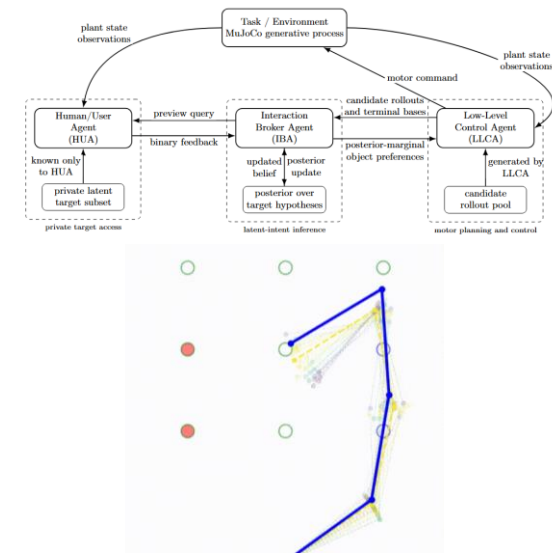
Modelled mouse point **and** click at trajectory level, with an active inference user model, including delay compensation.



Hierarchical closed-loops

Paterson, Stein, et al. (soon)

A high-DOF arm controlled via low-rate input, using hierarchical active inference control that partitions information gain and pragmatic value over timescales.



The four horsemen

Computation AIF is computationally intensive, lots of room to improve over naïve rollouts.

Implementation we are still building AIF software libraries, workflows or computational abstractions to build, reason about, and deploy interfaces.

Preference modelling Casting interaction problems in terms of preference priors is compelling and necessitates new ways to elicit, pool and validate them.

Uncertainty and prediction Representing uncertainty yields robustness, but users can struggle to understand uncertainty. Prediction is critical but can be unsettling for users. Opens a broad new frontier in UX.



An entreaty to rethink

1. View preference as a distribution, not a goal.
2. Use EFE as a fungible measure unifying exploitation and exploration.
3. Model joint action of dyads, not users and systems.
4. Shift Markov blankets to capture ephemeral agents.
5. Ask not what the signal implies, but what caused the signal.
6. Reason from the future to the now.

Less “design interfaces”. More “**engineer joint agents**”.

Predictive, probabilistic, preference-satisfying agents,
whose boundaries follow the action.